Rates

Teacher resource

Question 1

Apples cost $3.60 per kilogram.

What is the cost of 0.6 kilograms?

$0.60	$2.16	$3.00	$6.00
O	O	O	O

Answer: B
Skill: Students calculate with simple rates

Background information

Some secondary students have not yet developed a full appreciation of multiplication and division, the inverse relationship between the two operations and the application of these operations to problem solving. As a result, the use of multiplication and division to model and solve problems involving rates and scales is not obvious to all secondary students. While students may have developed these understandings for situations that involve familiar rates and whole number multipliers, they may not be able to generalise their thinking to situations involving less familiar rate situations and/or those which involve fractions and decimal multipliers.

Additional questions

1. Toni was charged $6.84 for some apples. The apples were $3.60 per kilogram. How much did they weigh?
2. A bag of apples cost $8.60. The apples weighed 3.4 kilograms. What was the cost per kilogram of these apples?

Question 2

Tracey drives 25 kilometres to work every day.

On Thursday her average speed was 75 kilometres per hour.
On Friday her average speed was 50 kilometres per hour.

How many more minutes did the trip take on Friday than Thursday?

[ ] minutes

Answer: 10 minutes
Skill: Students calculate with a familiar rate.

Additional questions

1. Tracey travels 80 kilometres at an average speed of 60 kilometres per hour. How many hours does the journey take?
2. A car travels at an average speed of 90 kilometres per hour. On average, how many metres are travelled each minute?

Question 3

An electronic photograph display shows 12 photographs per minute.
The display takes 3 hours to show all of Mary's photographs.
What speed, in photographs per minute, should be used to show all of Mary's photos in exactly 2 hours?

6	18	24	30
O	O	O	O

Answer: B
Skill: Students understand the relationships between speed, distance and time.

Additional questions

1. A car travels 50 kilometres at a particular speed. How far would the car travel in the same time if it travelled at twice the speed?
2. A car travels 50 kilometres in one hour. How long would it take the car to travel 25 kilometres if it travelled at twice the speed?

Question 4

A population of koalas has a death rate of 1 female koala every 5 days.
The same population has a death rate of 1 male koala every 4 days.

Which of the following statements about the overall death rate of koalas in the population is correct?

- O The death rate is higher than one per two days
- O The death rate is lower than one per two days
- O The death rate is higher than two per day
- O The death rate is lower than one per nine days

Answer: B
Skill: Students compare rates.

Additional questions

1. Approximately how many deaths of males and females would be expected in one month?
2. What percentage of koala deaths is due to deaths of males?

Student activity: Rates

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